Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody dependent cellular cytotoxicity; ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; ANOVA, analysis of variance; CHO, Chinese hamster ovary; Cpk, one-sided process capability index; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DO, dissolved oxygen; DoE, design of experiments; DS, drug substance; HCP, host cell protein; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G of the first subclass; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; pCO , dissolved carbon dioxide; PD, Process Development; PPQ, process performance qualification; PRESS, prediction sum of squares; QA, Quality Assurance; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SD, standard deviation; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; UPLC, ultra performance liquid chromatography; WCB, working cell bank.

1 Purpose and scope

This plan defines the process characterization studies for the production bioreactor (Step 3 of the A-Mab drug substance process). The step is a fed-batch mammalian cell culture operated at a commercial scale of 15,000 L, and it forms the glycan, charge variant and size variant critical quality attributes (CQAs) of the drug substance. Because no later step in the purification train alters the glycan distribution, the culture conditions established here carry through to release (CMC Biotech Working Group 2009).

The A-Mab drug substance process is being transferred from the development site in Cambridge to the commercial facility in Grafton under PTP-001, and Stage 1 process characterization is a prerequisite for the qualification runs at the receiving site (U.S. Food and Drug Administration 2011). The purpose of the studies described below is to establish, in a qualified scale-down model, the operating ranges that the commercial process will use and the evidence that supports them.

The scope of this plan is the production bioreactor from inoculation to the end of culture, covering the 9 process parameters listed in Table 6.1 over the ranges given there. Seed expansion up to and including the N-1 seed bioreactor is out of scope. Harvest and clarification (PCP-004) and each of the purification steps are out of scope as well, since they are covered by their own plans. Validation of the analytical methods is also out of scope, and is held in the method validation reports listed in Table 3.2. Confirmation at commercial scale is out of scope as well, since it belongs to process performance qualification (PPQ). This document states what will be done and the criteria by which the outcome will be judged, and it reports no data. The executed studies and their analysis will be reported in PCR-003.

2 Objectives

The studies have five objectives.

- (i) Quantify the effect of each parameter in Table 6.1 on each quality attribute in Table 4.1, across the ranges to be studied.
- (ii) Identify the parameters and the two-factor interactions that govern each attribute.
- (iii) Define a multivariate operating region within which every governed attribute is predicted to remain inside its acceptance criterion.
- (iv) Establish a proven acceptable range (PAR) for each parameter against each attribute it governs, and confirm that the normal operating range (NOR) lies inside it.
- (v) Provide the evidence needed to classify each parameter under SOP-4001 and to fix the control strategy that the qualification stage will operate to.

These objectives are addressed by two designed experiments (DoE) and one univariate assessment. The screening design serves objectives (i) and (ii), and the response-surface design serves (iii) and (iv). The classification called for in (v) is made in PCR-003, from the results of both designs together with the risk basis in RA-001.

3 Related documents

The documents this plan depends on, and the document it feeds, are listed in Table 3.1. RA-001 sets the scope of the studies, PCMP-001 sets the campaign acceptance criteria that this plan applies, and PCR-003 is the paired report in which the executed studies will be described.

Table 3.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures and validated analytical methods that govern execution and analysis are listed in Table 3.2, in which all reference numbers are placeholders for a controlled document system.

Table 3.2: Procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The production bioreactor is a fed-batch culture of a recombinant Chinese hamster ovary (CHO) cell line expressing A-Mab. The vessel is inoculated from the N-1 seed bioreactor at a target viable cell concentration, culture pH, temperature, dissolved oxygen (DO) and dissolved carbon dioxide (pCO) are controlled throughout the culture, and nutrient feeds are added on a fixed schedule. The culture is harvested at the assigned culture duration and passed to harvest and clarification (Step 4).

A-Mab is a humanized IgG1 monoclonal antibody produced on the Novacyte cell culture platform, and three prior products (X-Mab, Y-Mab and Z-Mab) have been manufactured on that platform in vessels of the same design and with the same basal medium and feed system. A-Mab itself has been produced for clinical supply and in engineering runs at the receiving site. This body of experience establishes the mechanisms that operate in the step, so the studies described in this plan are designed to confirm and to bound those mechanisms and not to discover them.

Platform experience supports four mechanistic expectations, and the design tests each of them.

- Culture pH and temperature set the intracellular processing of the N-glycan through their effect on glycosyltransferase and mannosidase activity, so both are expected to move high mannose and galactosylation.
- Dissolved carbon dioxide accumulates as cell density rises and lowers intracellular pH, and it is expected to act on cell specific productivity and on the charge variant distribution.
- Osmolality acts on growth and on cell specific productivity, and it is expected to interact with pCO because both are consequences of the feed and gassing strategy.
- Culture duration sets how long the product is held in the culture environment. Since aggregate and acidic variants accumulate late in the culture, duration is expected to be the dominant factor for both of those attributes.

The step forms impurities and removes none. Host cell protein (HCP) and residual DNA enter the harvest and are cleared downstream by Protein A capture, cation exchange and anion exchange. Those clearance claims are made in the reports for the steps concerned (PCR-005, PCR-007 and PCR-008). No viral clearance is claimed for the bioreactor. Viral safety rests on the modular claims of the low pH inactivation, anion exchange and virus filtration steps (PCR-006, PCR-008 and PCR-009), and the cumulative claim is consolidated in PCMR-001 (International Council for Harmonisation 2023a).

4.2 Quality attributes in scope

7 quality attributes are assigned to this step, and Table 4.1 gives their drug substance acceptance criteria together with the criticality assigned to each.

Table 4.1: Quality attributes formed at the production bioreactor, with drug substance acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

Criticality is assigned on a continuum and not as a binary split into critical and non-critical (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009), and the Tool #1 score shown in the table is the product of the impact of the attribute and the uncertainty around that impact. High mannose carries the highest of the 7 scores and is the only attribute of very high criticality. Afucosylation and galactosylation are of high criticality because A-Mab acts through antibody dependent cellular cytotoxicity (ADCC) and both attributes modulate Fc receptor binding, while aggregates are of high criticality on immunogenicity grounds. Acidic charge variants and residual DNA score low.

5 of the attributes will be measured as responses in the designed experiments (afucosylation, galactosylation, high mannose, acidic charge variants and high molecular weight aggregates). Host cell protein and residual DNA will be measured at harvest and reported, but they will not be modelled, because both are cleared by three downstream steps and the level leaving the bioreactor is not what constrains the drug substance (CMC Biotech Working Group 2009). Their acceptance criteria in Table 4.1 are drug substance criteria, and capability against them is assessed across the whole train in PCMR-001.

4.3 Risk-based prioritization of parameters

The scope of these studies follows the risk assessment in RA-001, in which every bioreactor parameter was ranked twice, once for its potential impact on the quality attributes and once for its potential interaction with other parameters (CMC Biotech Working Group 2009). Where no data or rationale were available to support an assessment, the parameter was ranked at the highest level. The ranking itself is held in RA-001 and is not repeated here.

9 parameters carry forward into this plan, of which 8 are assigned to multivariate study and 1 to univariate assessment. Parameters that RA-001 ranked low are not studied, and most of those are scale dependent. Agitation rate, gas flow rates and vessel pressure act on the culture through mass transfer, and they are fixed during qualification of the scale-down model so that the model matches commercial performance (Section 5.1).

The ranges to be studied are deliberately wider than the routine control ranges. Across the 5 factors of the designed experiments (Table 6.2), the studied range is between 1.6 and 2.5 times the width of the NOR. Wide ranges bound the knowledge space and leave room for later movement within the design space (CMC Biotech Working Group 2009), and they also improve the precision of the effect estimates, because each effect is then measured across a larger change in the factor. Each range was nevertheless set from platform experience so that the edges remain operable, and the edges are not intended to be edges of failure.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run in bench-scale stirred tank bioreactors qualified as a model of the commercial vessel under SOP-1001, with the culture operated under SOP-2003. No characterization run will be executed against an unqualified model.

The model matches the scale-independent variables and equalizes the effect of the scale-dependent ones (CMC Biotech Working Group 2009). Culture pH, temperature, DO, pCO , osmolality, initial viable cell concentration, feed schedule and culture duration will be operated at the commercial target values. The scale-dependent settings (agitation, gas flow rates, vessel pressure and working volume) will be established so that the bench vessel matches commercial performance in volumetric mass transfer coefficient, mixing time and carbon dioxide removal.

Qualification will compare replicate runs at set-point against the commercial and pilot scale record for the same conditions, covering growth, viability, titre and the 5 quality attributes measured as responses. The acceptance criterion is given in Section 7, and the qualification data will be reported in PCR-003.

The centre points of each design give a second and internal check on the model, because their replicate spread is the pure error term used in the lack of fit test (Section 5.5) and it also measures how reproducibly the model performs across the execution period.

5.2 Cell line, media and culture operation

One vial of the A-Mab working cell bank (WCB) will start the seed train for each block of runs. The seed train will be expanded to the N-1 stage under SOP-2003, and the production vessel will then be inoculated at the target initial viable cell concentration.

Basal medium and nutrient feeds will be prepared under SOP-2004, and a single lot of basal medium and a single lot of each feed will be used throughout the study, so that variation between lots does not enter the effect estimates. Medium and feeds are controlled to platform specification for identity and quality, and although basal medium concentration and nutrient feed volume are process parameters and appear in Table 6.1, both are held at their set-points in the designed runs described in Section 6.

Cultures will be fed on the platform schedule and harvested at the culture duration assigned to the run, and each run yields one clarified harvest on which the quality attributes are measured.

5.3 Analytical methods

The validated methods that will generate the study data are listed in Table 5.1.

Table 5.1: Validated analytical methods supporting the study.

Reference	Title	Type
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

The glycan attributes are measured by released N-glycan mapping, which reports afucosylation, galactosylation and high mannose from a single injection, while aggregates are measured by size exclusion chromatography (SEC-HPLC) and acidic charge variants by imaged capillary isoelectric focusing (icIEF). Host cell protein is measured by immunoassay and residual DNA by quantitative polymerase chain reaction (qPCR).

Each method is validated, and its validation report defines accuracy, precision, range and the reportable result. A summary of method performance will be given in PCR-003 (Appendix D). Sample handling and storage follow the method procedures listed in Table 3.2, and harvest samples for the glycan and size variant assays will be frozen and analysed in a single analytical campaign per design, so that analytical drift across the execution period does not confound the run order.

5.4 Sampling plan

Two kinds of sample will be taken. Daily process samples support the conduct of the culture and are used for viable cell concentration, viability, metabolites, osmolality and pCO , while a single harvest sample from each run supports the quality attribute measurements.

Every run in the matrices of Appendix A and Appendix B yields one harvest sample. The screening design includes 3 centre point runs and the response-surface design includes 4, and these runs are executed at identical settings and distributed across the execution period, so that they measure reproducibility over the whole campaign and not the reproducibility of a single day. Retained harvest material will be stored under the conditions given in the method procedures, so that a disputed result can be re-tested.

5.5 Statistical methods

Analysis will follow SOP-1002 and will be carried out on coded factors, where the set-point is 0 and the edges of the studied range are the coded levels -1 and $+1$. Coding makes the effect estimates comparable across factors that have different units and different range widths.

The screening design will be analysed by ordinary least squares (OLS), with all main effects and all two-factor interactions in the model. The effect of a term is reported as twice its coded coefficient, so that an effect is the change in the response across the studied range. Terms will be judged at a significance level of 0.05.

The response-surface design will be analysed as a full quadratic model in its factors, with main effects, two-factor interactions and pure quadratic terms. Adequacy will be judged from the coefficient of determination and its adjusted and predicted forms, from the overall F test, and from an analysis of variance (ANOVA) that partitions the residual into lack of fit and pure error. The predicted coefficient is computed from the prediction sum of squares (PRESS). Residual diagnostics will be inspected for structure that the model has not captured (residuals against the predicted values, a normal quantile plot of the residuals, and the predicted values against the measured ones).

A factor with no significant effect on a response is a result and not a gap in the study. Such factors will be reported with their effect estimate and its standard error. They will be kept in the knowledge space as evidence that the response is robust to them across the range studied.

Capability at commercial scale will be estimated by Monte Carlo simulation of 2,000 batches, with each parameter drawn inside its NOR and with the residual variability of the fitted model added. A one-sided capability index (Cpk) will then be computed for each modelled attribute against the limit that governs it.

6 Study design

6.1 Factors, ranges and study type

The parameters to be studied are listed in Table 6.1, with their set-points, the ranges to be studied, the normal operating ranges and the type of study each will receive.

Table 6.1: Process parameters in scope, with set-points, ranges to be studied, normal operating ranges and the type of study each will receive.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
Dissolved CO2 (pCO2)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% sat	50	30–70	40–60	multivariate
Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	X	1.2	0.8–1.6	1–1.4	multivariate
Nutrient feed-1 volume	% WV	12	9–15	10–14	multivariate

Of the 9 parameters in scope, 8 are assigned to multivariate study and 1 to univariate assessment. The designed experiments described below carry the 5 factors that RA-001 ranked highest for impact and for interaction. Those factors, and the code letters used for them in the design matrices and in the analysis, are given in Table 6.2.

Table 6.2: Coded factor legend for the designed experiments.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO2 (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

Dissolved oxygen, basal medium concentration and nutrient feed volume are held at their set-points in the designs described here, and are assessed univariately (Table 6.1). Their ranges rest on the platform characterization carried out on the prior products, in vessels of the same design and with the same medium system. If the screening results show that this platform claim does not transfer to A-Mab, the three parameters will be added to a further design under a numbered amendment to this plan.

The range of each factor was chosen for a stated reason. Culture pH is studied over 6.6–7.1, which is the widest range relative to its control range. The pH set-point is the principal handle on the glycan distribution, and the operating region needs room around it. Culture temperature is studied over 34–36 °C, the band that the commercial vessel can hold with its jacket at full duty. Dissolved carbon dioxide is studied over 40–160 mmHg, which spans what the commercial vessel accumulates at high cell density (stripping is less efficient at scale than at bench scale). Osmolality is studied over 360–440 mOsm, the range the feed strategy can produce, and culture duration over 15–19 days, the range that harvest scheduling can accommodate.

6.2 Screening design

The screening design identifies which factors and which two-factor interactions matter, and it is not the predictive model.

A two-level fractional factorial in the 5 screening factors will be executed and augmented with centre points, giving 19 runs in total (16 factorial runs and 3 centre points). The fraction is a half fraction of resolution V, so every main effect and every two-factor interaction is estimable and each is aliased only with interactions of three factors or more, while the centre points give an independent estimate of pure error and a test for curvature. Runs will be executed in randomized order under SOP-1002, and the planned matrix is given in Appendix A.

The screening analysis will report each effect with its standard error and p-value. Effect magnitudes from a near-saturated screening model estimate direction and size, and they will be refined by the response-surface model (RSM). No operating region and no proven acceptable range will be claimed from the screening design.

6.3 Response-surface design

The response-surface design is the predictive model and the basis of the operating region.

A face-centred central composite design will be executed in the 4 factors carried forward from screening, giving 28 runs made up of 16 factorial runs, 8 axial runs and 4 centre points. The axial points sit on the faces of the cube, at the coded levels -1 and $+1$, so that no run is executed outside the range already covered by screening. That choice costs some rotatability, and it buys a design in which every setting is one the commercial vessel can hold.

Prior knowledge indicates that the factors carried forward will be culture pH, culture temperature, culture duration and dissolved carbon dioxide, and the matrix in Appendix B is written for that set. If screening identifies a different set, the matrix will be re-issued as an amendment before the response-surface runs are executed.

The quadratic terms allow curvature in each factor and the cross terms allow a response to depend on combinations of factors, which is why the operating space is described as a multivariate region and not as a set of independent ranges (International Council for Harmonisation 2009).

6.4 Univariate assessment

Initial viable cell concentration will be assessed one factor at a time over 0.5–1.5 e6 cells/mL, with all other parameters held at set-point. It is not carried into the designed experiments, because it acts on the culture through the length of the growth phase, and that effect is already spanned by culture duration. It is also set at inoculation and controlled by the N-1 step, so its variability at commercial scale is small.

The univariate runs will be compared against the centre point runs of the screening design, using the pure error estimate from those centre points as the reference variance.

7 Acceptance and decision criteria

The study passes when every criterion set out below is met. Each criterion is stated before execution and will be assessed as written.

7.1 Qualification of the scale-down model

The model is qualified when the performance of replicate runs at set-point is not statistically different from the commercial scale record for the same conditions at a significance level of 0.05, the comparison covering growth, viability, titre and the 5 modelled quality attributes. If any of them differs, the difference will be investigated and either the model will be modified or the claim made for it will be narrowed, in both cases before characterization runs begin (U.S. Food and Drug Administration 2011).

7.2 Reproducibility of the centre points

The percent coefficient of variation of each response across the centre points will be reported for both designs, and a response whose centre point variation exceeds the intermediate precision stated in its validation report will be investigated before the design is analysed. Centre point variation is also the pure error term, so an inflated value widens the confidence limits on every effect and weakens the lack of fit test. This criterion is therefore assessed before any effect is interpreted.

7.3 Adequacy of the fitted models

A response-surface model will be treated as adequate for prediction when three conditions hold. The overall F test is significant at 0.05, and the lack of fit F test is not significant against pure error at the same level. The predicted coefficient of determination is also close to the adjusted coefficient, which shows that the fit is not carried by a small number of high leverage runs. A model that fails any of these conditions will be reported as characterized but not predictive, the ranges for the affected attribute will then be claimed from the univariate analysis described in Section 8, and the shortfall will be stated in PCR-003.

7.4 Definition of the operating region

The operating region will be declared as the set of factor combinations over which the fitted models predict every governed attribute to lie inside its acceptance criterion (Table 4.1). It will be reported with the factor planes that describe it, with its worst case corner named, and with the NOR located inside it. The region is a mean prediction region, so reliability at its edge is lower than at its centre, and the capability analysis and the PAR analysis are what bound that (CMC Biotech Working Group 2009). Movement inside the declared region is not a regulatory change (International Council for Harmonisation 2009), although it remains subject to the change management system of the pharmaceutical quality system (International Council for Harmonisation 2008).

7.5 Capability at commercial scale

Each modelled attribute will be assessed for capability against the limit that governs it, using the Monte Carlo simulation described in Section 5.5, and the minimum acceptable capability index is the value fixed in PCMP-001. An attribute that falls below that value will be reported with the margin it does achieve, and the operating region will be narrowed until the criterion is met.

7.6 Classification of parameters

Each parameter will be classified under SOP-4001 from two inputs: the size of its demonstrated effect on the quality attributes, and the risk that normal operation carries it outside the declared region. A parameter with no significant effect is still classified, and the evidence of robustness is recorded with it. Classification is an output of PCR-003 and is not anticipated in this plan.

7.7 Handling of deviations

Any departure from this plan will be recorded as a deviation, investigated for root cause and dispositioned before the affected data are used. Where the investigation shows that a deviation did not move a factor outside its studied range and did not affect the measured responses, the runs will be retained and the argument recorded. Where a deviation invalidates a design for the purpose of defining the operating region, that design will be re-executed in full on requalified material. The first execution will be retained as a superseded dataset and reported as such, and only the requalified execution will be analysed for the operating region.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of one parameter over which a quality attribute stays inside its acceptance criterion while the other parameters are treated in a stated way, and this section fixes how those ranges will be determined before any data exist.

The acceptance basis is the drug substance criterion for each attribute (Table 4.1). The bioreactor sets no viral clearance attribute, so no step level clearance floor has to be back-calculated here. That calculation applies to the low pH inactivation, anion exchange and virus filtration steps, whose cumulative claim is consolidated in PCMR-001 (International Council for Harmonisation 2023a).

Two analyses will be run for each governed attribute and each factor of the response-surface model. The first holds the other factors at their set-points, varies the parameter across its full characterization range, and evaluates the fitted model on a grid across that range. The PAR is then the contiguous part of the grid that contains the set-point and over which the predicted attribute stays inside acceptance.

The second analysis varies the other factors within their normal operating ranges. At each grid point, 2,000 draws are taken from the fitted model, with the other factors sampled inside their NORs and with the residual variability of the model added. The PAR is the contiguous part of the grid over which the predictive interval of the attribute stays inside acceptance. This is a robustness criterion, so the second PAR can only be narrower than the first or equal to it. The simulation is seeded, which makes both analyses reproducible from the fitted model and the parameter definitions alone.

The decision rule is the same for both analyses. Where the acceptable range covers the whole characterization range, that whole studied range will be reported as proven acceptable for the attribute in question. Where it does not, the binding attribute will be named, together with the margin between the PAR and the NOR. A parameter whose PAR does not contain its NOR fails the criterion, and the NOR will then be narrowed in PCR-003.

Results will be reported in two forms. A table will give, for each attribute and each parameter, the characterization range and both proven acceptable ranges. A plot will be given for each governed attribute against the parameter with the largest main effect on it. The plot carries the parameter on the horizontal axis and the attribute on the vertical axis, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and NOR marked. The two analyses appear as the two panels of that plot.

The relationship between these ranges and the operating region will be stated explicitly in the report. A PAR is univariate and is conditioned on the fitted model, on the characterization ranges and on how the other factors are treated in each analysis, whereas the operating region defined in Section 7 is multivariate. Where the whole characterization range of a parameter is proven acceptable, the binding constraint on operation is the multivariate corner of the region and not any univariate range.

9 Data management and integrity

Process data will be captured by the bioreactor control system and its validated historian, analytical results will be captured in the validated chromatography data system and the laboratory information system, and manual entries will be recorded in the validated electronic laboratory notebook.

All records will meet ALCOA+ expectations, so that entries are attributable, legible, contemporaneous, original and accurate, and the record as a whole is complete, consistent, enduring and available. Audit trails will be enabled and reviewed.

The analysis datasets, the scripts that produce the tables and figures, and the fitted models will be held under version control, and a second statistician will reproduce the analysis independently before PCR-003 is issued, as required by SOP-1002.

10 Roles and responsibilities

Responsibilities are assigned across the functions involved in Table 10.1. Process Development owns the plan, its execution and the report, while Quality Assurance approves both documents and dispositions any deviation raised during execution.

Table 10.1: Functions and their responsibilities for the study.

Function	Responsibility
Process Development	Owns this plan. Designs, executes and analyses the studies. Authors PCR-003.
Manufacturing Science and Technology	Owns qualification of the scale-down model and the transfer package to the receiving site.
Analytical Development	Owns the validated methods, sample analysis and release of analytical data.
Statistics	Reviews the designs before execution and reproduces the analysis independently.
Commercial Manufacturing, Grafton	Reviews the ranges for compatibility with the commercial vessel and its control system.
Quality Assurance	Approves this plan and PCR-003. Dispositions deviations.

11 Deliverables and schedule

The deliverable of this plan is PCR-003, the process characterization report for the production bioreactor, which will contain the qualification of the scale-down model, the executed designs, the fitted models, the operating region, the proven acceptable ranges, the capability assessment and the parameter classification.

Execution follows the phase sequence given in Table 11.1, and each phase gates the next. No dates are fixed in this plan, because the schedule for the campaign is held in PTP-001.

Table 11.1: Phase sequence and the gate between phases.

Phase	Content	Gate to the next phase
Model qualification	Replicate runs at set-point against the commercial record	Model qualified under SOP-1001
Screening	Fractional factorial with centre points	Active factors identified and the response-surface matrix confirmed
Response surface	Face-centred central composite design	Adequate models for the governed attributes
Univariate assessment	Initial viable cell concentration across its range	Data complete and reviewed
Analysis and reporting	Operating region, proven acceptable ranges, capability, classification	PCR-003 approved

12 Risks and assumptions

The main assumption is that the scale-down model represents the commercial vessel for the attributes studied. The risk is that it does not, in which case the operating region derived from it would not transfer. This is managed by qualifying the model before execution, by the criterion in Section 7, and by stating any residual limitation in PCR-003.

A single lot of basal medium and a single lot of each feed will be used. This keeps lot variability out of the effect estimates, but it also means that the study says nothing about that variability. Variation between lots is controlled by the material specifications under SOP-2004 and is covered by the platform record.

Assay variability sets a floor on the smallest effect the study can resolve, and high mannose is the attribute most exposed to it, because its acceptance range is narrow relative to the precision of the glycan method. The centre point replicates measure that floor directly, and effects will be reported against it.

The response-surface design assumes that screening identifies the factors that matter, and if screening identifies a different set the matrix in Appendix B will be re-issued before execution. That is an amendment to this plan and not a deviation from it.

The operating region will be defined on predicted mean values, so reliability at the edge of the region is lower than at its centre, and the capability and PAR analyses exist to bound that. This plan does not claim that the edges of the region will be confirmed at commercial scale. Confirmation at scale belongs to PPQ, and the operating region is not a substitute for it.

Material supply is a schedule risk, because each block of runs needs an N-1 seed culture and the runs in a block are executed together. Loss of a seed train delays a block, and the affected runs will then be repeated in a later block with their own centre points.

13 Approvals

This plan takes effect when it has been approved by all of the functions listed in Table 1, and approval signifies agreement with the objectives, the study design and the acceptance criteria stated here. Any change after approval will be issued as a numbered amendment and approved by the same functions.

14 Appendices A and B — Planned design matrices

The matrices below are planned settings and carry no responses, because none have been measured. The code letters are defined in Table 6.2.

14.1 Appendix A — Planned screening design matrix

The coded matrix is given first, then the same runs in natural units. Run order will be randomized at execution and recorded in the study record.

Table 14.1: Planned screening design, coded settings.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Table 14.2: Planned screening design, settings in natural units.

Run	Type	Culture pH	Culture temperature	Dissolved CO2 (pCO2)	Osmolality	Culture duration
1	factorial	6.6	34	40	360	19
2	factorial	6.6	34	40	440	15
3	factorial	6.6	34	160	360	15
4	factorial	6.6	34	160	440	19
5	factorial	6.6	36	40	360	15
6	factorial	6.6	36	40	440	19
7	factorial	6.6	36	160	360	19
8	factorial	6.6	36	160	440	15
9	factorial	7.1	34	40	360	15
10	factorial	7.1	34	40	440	19
11	factorial	7.1	34	160	360	19
12	factorial	7.1	34	160	440	15
13	factorial	7.1	36	40	360	19
14	factorial	7.1	36	40	440	15
15	factorial	7.1	36	160	360	15
16	factorial	7.1	36	160	440	19
17	center	6.85	35	100	400	17
18	center	6.85	35	100	400	17
19	center	6.85	35	100	400	17

14.2 Appendix B — Planned response-surface design matrix

The design is written for the factors expected to be carried forward from screening, and it will be re-issued as an amendment if screening identifies a different set.

Table 14.3: Planned response-surface design, coded settings.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1

Run	Type	A	B	E	C
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 14.4: Planned response-surface design, settings in natural units.

Run	Type	Culture pH	Culture temperature	Culture duration	Dissolved CO2 (pCO2)
1	factorial	6.6	34	15	40
2	factorial	6.6	34	15	160
3	factorial	6.6	34	19	40
4	factorial	6.6	34	19	160
5	factorial	6.6	36	15	40
6	factorial	6.6	36	15	160
7	factorial	6.6	36	19	40
8	factorial	6.6	36	19	160
9	factorial	7.1	34	15	40
10	factorial	7.1	34	15	160
11	factorial	7.1	34	19	40
12	factorial	7.1	34	19	160
13	factorial	7.1	36	15	40
14	factorial	7.1	36	15	160
15	factorial	7.1	36	19	40
16	factorial	7.1	36	19	160
17	axial	6.6	35	17	100
18	axial	7.1	35	17	100
19	axial	6.85	34	17	100
20	axial	6.85	36	17	100
21	axial	6.85	35	15	100
22	axial	6.85	35	19	100
23	axial	6.85	35	17	40
24	axial	6.85	35	17	160
25	center	6.85	35	17	100

Run	Type	Culture pH	Culture temperature	Culture duration	Dissolved CO2 (pCO2)
26	center	6.85	35	17	100
27	center	6.85	35	17	100
28	center	6.85	35	17	100

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.